

Generative AI (GenAI)

Definition: A broad category of Artificial Intelligence focused on creating new content. Unlike traditional AI that classifies or predicts data, GenAI uses patterns from existing data to produce original text, images, code, or audio.

GANs (Generative Adversarial Networks)

Definition: A specific architecture consisting of two neural networks—the **Generator** (which creates data) and the **Discriminator** (which evaluates it). They "fight" each other until the Generator produces content indistinguishable from reality.

LLMs (Large Language Models)

Definition: Advanced GenAI models trained on massive datasets of human language. They predict the next word in a sequence, allowing them to engage in conversation, summarize text, and translate languages..

Vector

Definition: In AI, data (like a word or image) is converted into a list of numbers called a "Vector." These numbers represent the **semantic meaning** of the data in a multi-dimensional space.

VectorDB (Vector Database)

Definition: A specialized database designed to store and index vectors. Unlike traditional databases that search for exact keywords, VectorDBs search for "nearest neighbors"—finding data with similar meanings.

FAISS (Facebook AI Similarity Search)

Definition: An open-source library developed by Meta that allows for extremely fast searching of dense vectors. It is essentially an engine used to find similar items within a massive collection of vectors.

RAG (Retrieval-Augmented Generation)

Definition: A framework that provides an LLM with access to external, real-time, or private data. Instead of relying solely on its training, the model "retrieves" a relevant document first and then "generates" an answer based on that factsheet.

LangChain

Definition: An open-source framework designed to simplify the creation of LLM applications. It acts as the "orchestrator" or "glue" that connects LLMs with other components like VectorDBs, APIs, and memory.